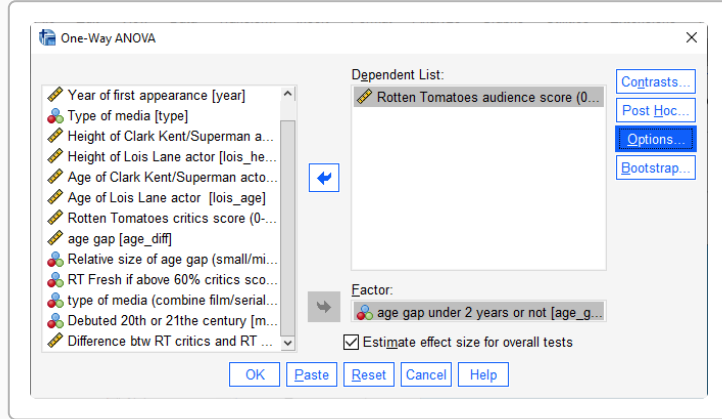


Bar Graphs to go with Between Groups ANOVA

Analyze → Graphs → Legacy Dialogs → Bar

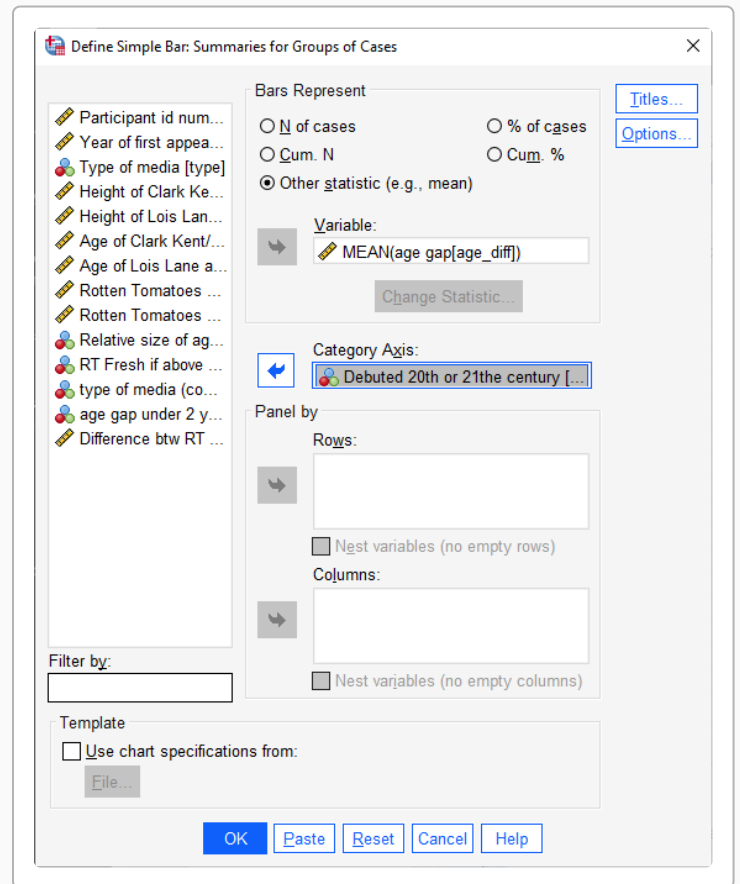
As for the BG ANOVA...

- One variable (modern) holds the group membership
- One variable (age_diff) holds the DV



#1 Select type of chart

- Click on “Simple” icon
- Click the “Summaries for groups of Cases” radio button
- Click “Define”



#2 Select the DV

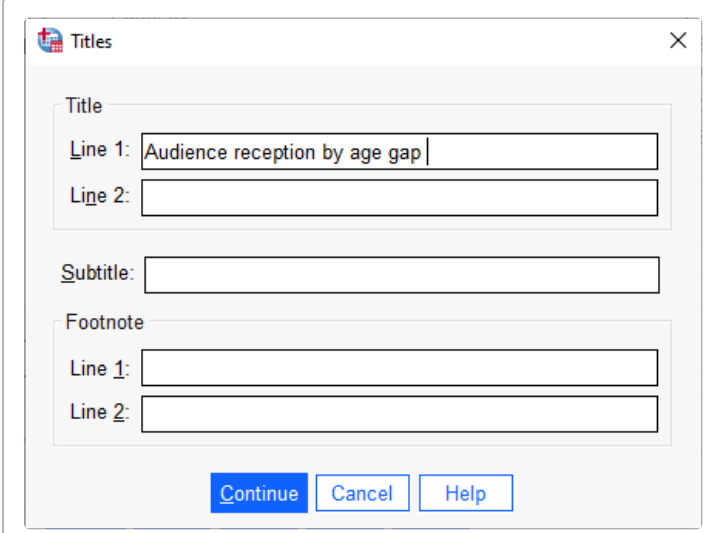
- Highlight the quantitative DV (age difference)
- Click the **“Other Statistic (e.g., mean)”** radio button — this is critical! The default is “N of cases” which would give you counts, not means
- Click the arrow to move the DV into the “Variable” window

#3 Select the IV

- Click the categorical IV (modern)
- Click the arrow to move the IV into the “Category Axis” window

#4 Write the Title

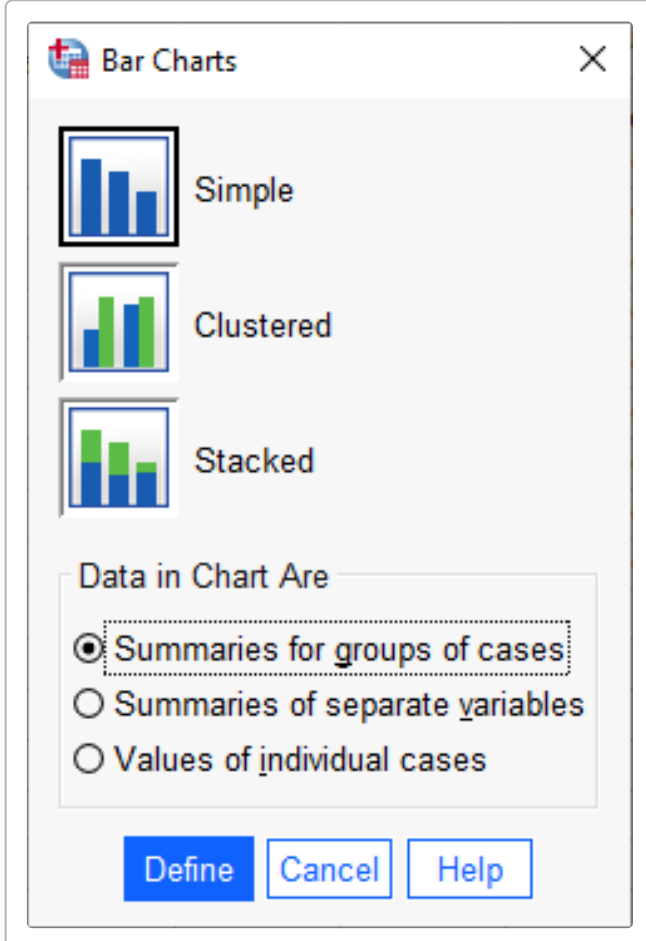
- Click on “Titles”
- Type titles, subtitles & footnotes as desired
- Click “Continue”



The "Titles" dialog box in SPSS is shown. It has a title bar with a red cross icon and the word "Titles". The dialog contains several input fields: "Line 1:" with the text "Audience reception by age gap", "Line 2:" (empty), "Subtitle:" (empty), and a "Footnote" section with "Line 1:" and "Line 2:" (both empty). At the bottom are three buttons: "Continue" (blue), "Cancel" (white with blue border), and "Help" (white with blue border).

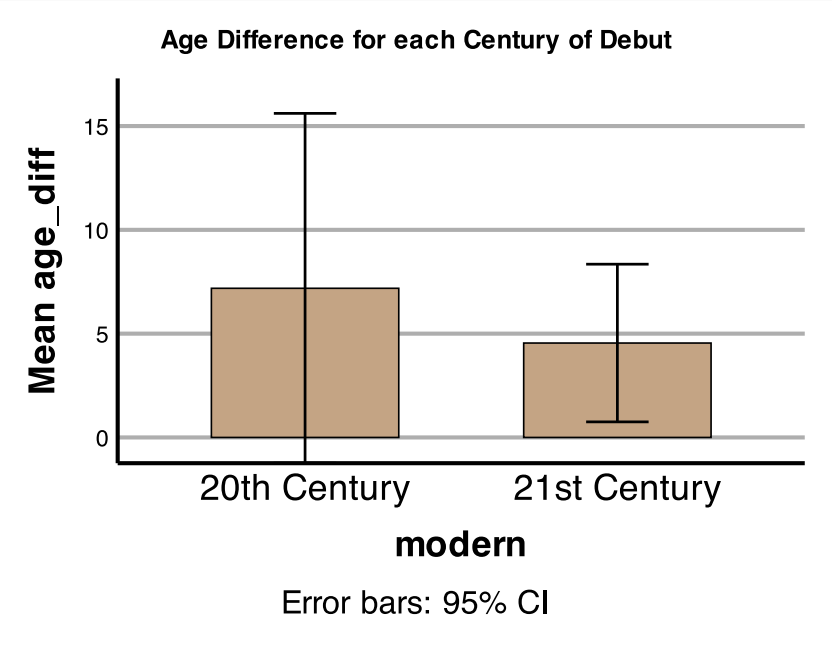
#5 Select the Whiskers

- Click Options
- Check "Display error bars"
- Check "Standard Error" or "Confidence intervals"
- Click Continue



The "Bar Charts" dialog box in SPSS is shown. It has a title bar with a red cross icon and the text "Bar Charts". The dialog contains three bar chart icons: "Simple" (blue bars), "Clustered" (green bars), and "Stacked" (green bars on top of blue bars). The "Simple" option is selected with a black border. Below the icons is a section titled "Data in Chart Are" with three radio button options: "Summaries for groups of cases" (selected), "Summaries of separate variables", and "Values of individual cases". At the bottom are three buttons: "Define" (blue), "Cancel" (white with blue border), and "Help" (white with blue border).

SPSS Output



#6 Interpreting the Graph

Bar Heights (Group Means): Each bar represents the average value of the dependent variable for that group. In this example, the bars show the mean age difference for Superman portrayals from the 20th Century versus the 21st Century.

Error Bars (95% Confidence Intervals): The lines extending above and below each bar represent the 95% confidence interval around each group mean. These intervals indicate the range where we can be 95% confident the true population mean lies.

Comparing Groups: When error bars from different groups do not overlap, this suggests the group means are significantly different from each other at the 0.05 significance level. If error bars do overlap, the differences may not be statistically significant.

Using the Graph: Right-click the graph in the SPSS output window and select “Copy” to put the graph into a Word or other file.